Homework Assignment 1

Math 292, Spring 2026

Rutgers University

Jeffrey Lu

March 3, 2026

Question 6.8

Let $a, b, c \in \mathbb{Z}$. For sake of contradiction, we assume $a^2 + b^2 = c^2$, and both a and b are odd. Then, by definition of odd, we have an integer x such that $a = 2x + 1$ and an integer y such that $b = 2y + 1$.

Now, we substitute our new definition of a and b into the original equation, giving us

$$\begin{aligned} a^2 + b^2 &= (2x+1)^2 + (2y+1)^2 \\ &= 4x^2 + 4x + 4y^2 + 4y + 2 \\ &= 2(2x^2 + 2x + 2y^2 + 2y + 1). \end{aligned}$$

We let $w = 2x^2 + 2x + 2y^2 + 2y + 1$, and w must be an integer because it is the linear combination of integers. Then, we have that

$$a^2 + b^2 = c^2 = 2w.$$

So, by definition of even, c^2 must be even, and since only the square of even integers can produce another even integer, c must also be even. Thus, by definition of even, there is an integer k such that $c = 2k$.

Then, the square must be

$$c^2 = (2k)^2 = 4k^2.$$

We let $z = k^2$, then z must be an integer because the product of integers is still an integer. Now, we have $c^2 = 4z$. Hence $4 \mid c^2$. We return to $a^2 + b^2$,

$$\begin{aligned} a^2 + b^2 &= 4x^2 + 4x + 4y^2 + 4y + 2 \\ &= 4(x^2 + x + y^2 + y) + 2. \end{aligned}$$

We let $n = x^2 + x + y^2 + y$, where n must be an integer because it is the linear combination of integers. This gives us

$$\begin{aligned} & a^2 + b^2 = 4n + 2 \\ \implies & c^2 - 2 = 4n \\ \implies & c^2 = 2 mod(4) \\ \implies & 4 \nmid c^2 \end{aligned}$$

This is a contradictory to when we concluded that $4 \mid c^2$. Therefore, if $a^2 + b^2 = c^2$, then a or b have to be even. $\square$

Question 6.10

For sake of contradiction, we assume that there exists integers a and b, such that $21a + 30b = 1$. Then factor out a 3

$$3(7a + 10b) = 1.$$

Let $k = 7a + 10b$, then k must be an integer since it's the linear combination of integers. Thus, we have

$$\begin{aligned} 3k &= 1 \\ k &= \frac{1}{3}. \end{aligned}$$

This is an contradiction because we concluded that k was an integer earlier. Therefore, we conclude that there exists no integers a and b for which $21a + 30b = 1$.

Question 6.12

Let x be a positive rational number. Then, we can rewrite x as the fraction of two positive integers p and q since x is positive.

$$x = \frac{p}{q}.$$

Next, we define $y = \frac{x}{2}$, such that $y = \frac{p}{2q}$. Since $x > 0$, dividing x by 2 yields a strictly smaller positive number, so $0 < \frac{x}{2} < x$. Let $w = 2q$, then w is an integer because it's the linear combination of integers. Then, we have

$$y = \frac{p}{w}.$$

Therefore, y is a positive rational number such that $y < x$.

Question 7.4

Suppose we have an even integer a. By definition of even, we have an integer x such that $a = 2x$. Then, we have that

$$\begin{aligned} a^2 + 4a + 5 &= (2x)^2 + 4(2x) + 5 \\ &= 4x^2 + 8x + 5 \\ &= 2(2x^2 + 4x + 2) + 1. \end{aligned}$$

Let $w = 2x^2 + 4x + 2$, then w must be an integer since it's the linear combination of integers. Hence, we have $a^2 + 4a + 5 = 2w + 1$. Thus, by definition of odd, we have that $a^2 + 4a + 5$ is odd. Then, we conclude that if a is even, then $a^2 + 4a + 5$ is odd.

Now we prove the converse using the contrapositive. Suppose a is an odd integer. Then, by definition of odd, we have integer y such that $a = 2y + 1$. Thus, we have that

$$\begin{aligned} a^2 + 4a + 5 &= (2y + 1)^2 + 4(2y + 1) + 5 \\ &= 4y^2 + 12y + 10 \\ &= 2(2y^2 + 6y + 5). \end{aligned}$$

We let $z = 2y^2 + 6y + 5$, and z must be an integer since it's the linear combination of integers. Hence, we have $a^2 + 4a + 5 = 2z$. Thus, by definition of even, we have that $a^2 + 4a + 5$ is even. We conclude that if $a^2 + 4a + 5$ is odd, then a is even. Therefore, we have that given an integer a, then $a^2 + 4a + 5$ is odd if and only if a is even.